Ansible Challenge -01

Create ci pipeline that will deploy and configure these VMs using terraform and ansible according following requirements

1. Deploy 2 virtual machines using terraform.
2. first vm on Amazon linux, hostname: c8.local
3. second vm on ubuntu 21.04, hostname: u21.local
4. As a result of terraform execution, dynamically create inventory for ansible
5. c8.local should be in the frontend group
6. u21.local should be in the backend group
7. Create ansible playbook for c8.local and u21.local

- for Linux OS playbook should apply the following changes
- selinux: disable
- firewalld: disable
- for frontend playbook group should install and configure nginx
- nginx configuration should do proxying from port 80 on port 19999 to the
- backend group
- for the backend group, the playbook must install the Netdata application from the official repositories and run on port 19999.

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```
provider "aws" {
  region = "us-east-1"
}

resource "aws_instance" "frontend" {
  ami           = "ami-0f4b77a23eac2378a"
  instance_type = "t2.micro"
  key_name      = "Linux-key"

  tags = {
    Name = "c8.local"
  }
}

resource "aws_instance" "backend" {
  ami           = "ami-0fc5d935ebf8bc3bc"
  instance_type = "t2.micro"
  key_name      = "Linux-key"

  tags = {
    Name = "u21.local"
  }
}
```

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- output.tf > output "backend_ip"

```
output "frontend_ip" {
  value = aws_instance.frontend.public_ip
}

output "backend_ip" {
  value = aws_instance.backend.public_ip
}
```

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```
aws_instance.frontend: Still creating... [00m30s elapsed]
aws_instance.backend: Still creating... [00m30s elapsed]
aws_instance.frontend: Creation complete after 37s [id=i-075dab38699b256f8]
aws_instance.backend: Still creating... [00m40s elapsed]
aws_instance.backend: Creation complete after 47s [id=i-02432a457029191d4]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

backend_ip = "34.207.150.171"
frontend_ip = "54.82.249.44"
PS C:\Users\Vigaas\Desktop\Testing\Project>
```

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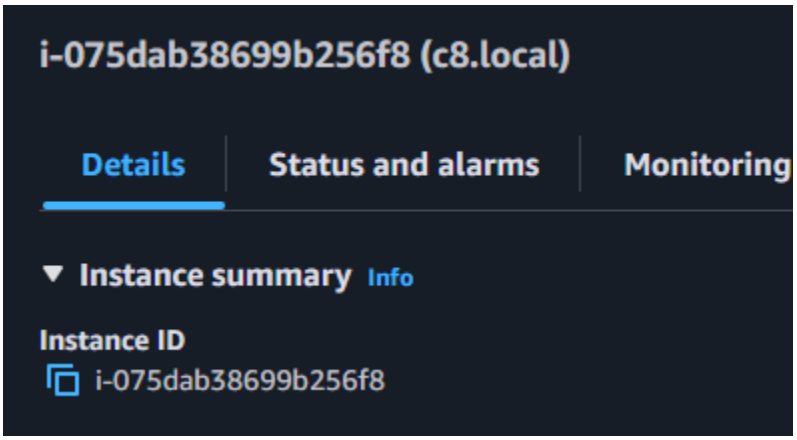

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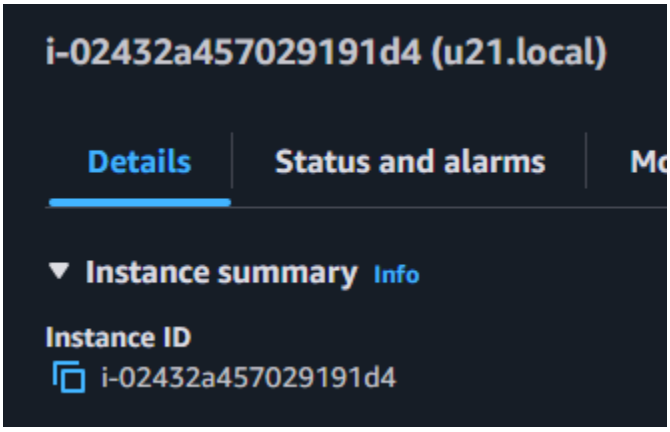

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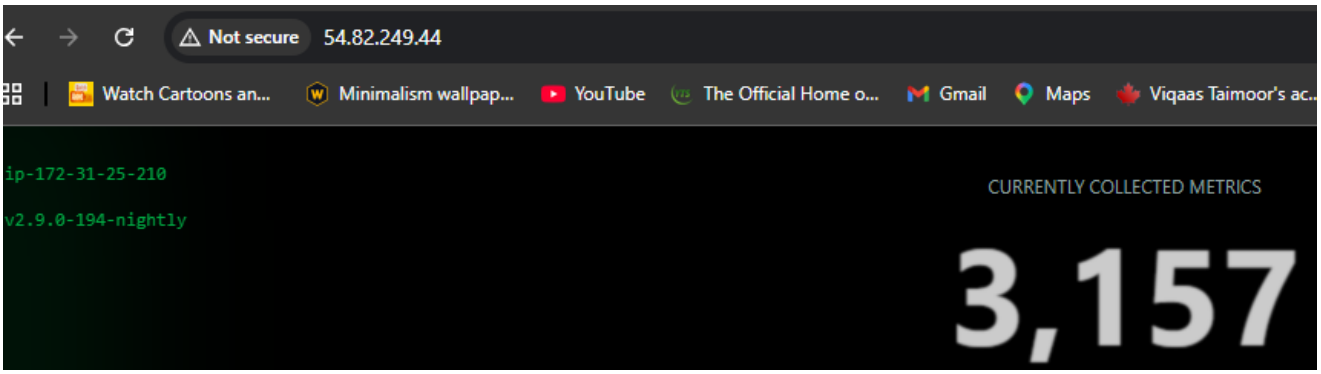
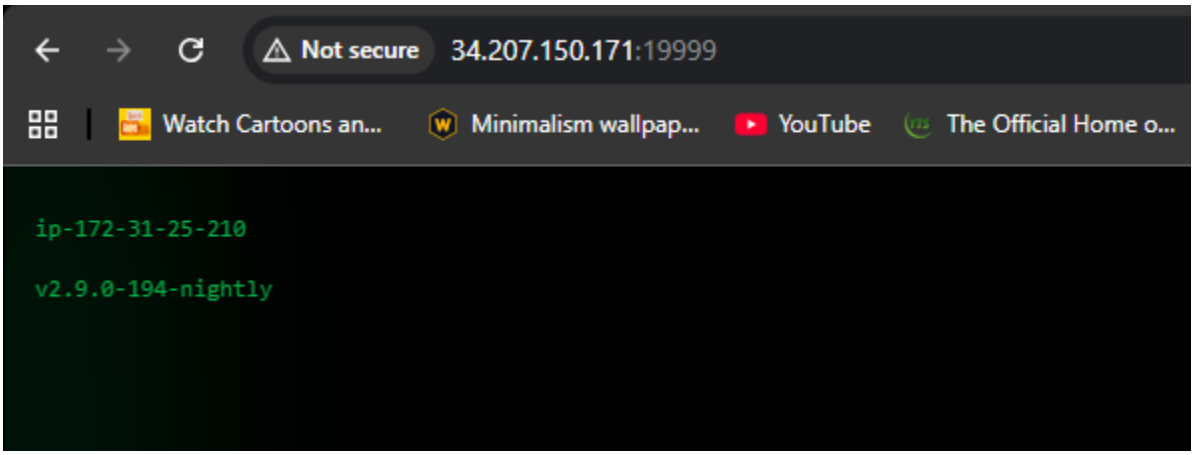
- ```
  root@408430b38a88:/var/jenkins_home/workspace/Ansible# ls
  inventory.ini  playbook.yml
  root@408430b38a88:/var/jenkins_home/workspace/Ansible# cat inventory.ini
  [frontend]
  c8.local ansible_host=54.82.249.44 ansible_user=ec2-user ansible_ssh_private_key_file=/var/jenkins_home/Linux-key.pem
  [backend]
  u21.local ansible_host=34.207.150.171 ansible_user=ubuntu ansible_ssh_private_key_file=/var/jenkins_home/Linux-key.pem
  root@408430b38a88:/var/jenkins_home/workspace/Ansible#
  ```

- 

- 


- ```
  - hosts: all
    become: yes
    tasks:

      - name: Disable SELinux (Amazon Linux)
        shell: |
          setenforce 0 || true
          sed -i 's/^SELINUX=.*/SELINUX=disabled/' /etc/selinux/config || true
        when: ansible_distribution == "Amazon"

      - name: Disable UFW (Ubuntu)
        service:
          name: ufw
          state: stopped
          enabled: no
        when: ansible_distribution == "Ubuntu"
        ignore_errors: yes
  ```

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```
- hosts: frontend
  become: yes
  tasks:

    - name: Install nginx
      dnf:
        name: nginx
        state: present

    - name: Start nginx
      service:
        name: nginx
        state: started
        enabled: yes

    - name: Configure nginx proxy
      copy:
        dest: /etc/nginx/nginx.conf
        content: |
          events {}
          http {
            server {
              listen 80;
              location / {
                proxy_pass http://{{ hostvars[groups['backend'][0]].ansible_host }}:19999;
              }
            }
          }

    - name: Restart nginx
      service:
        name: nginx
        state: restarted


# ========================
# BACKEND (Ubuntu)
# ========================
- hosts: backend
  become: yes
  tasks:

    - name: Install Netdata
      shell: curl -Ls https://my-netdata.io/kickstart.sh | bash

    - name: Start Netdata
      service:
        name: netdata
        state: started
        enabled: yes
```

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```
pipeline {
  agent any

  environment {
    ANSIBLE_HOST_KEY_CHECKING = "False"
    FRONTEND_IP = "54.82.249.44"
    BACKEND_IP  = "34.207.150.171"
  }

  stages {

    stage('Create Inventory') {
      steps {
        sh """
        echo "[frontend]" > inventory.ini
        echo "c8.local ansible_host=$FRONTEND_IP ansible_user=ec2-user ansible_ssh_private_key_file=/var/jenkins_home/Linux-key.pem ansible_python_interpreter=/usr/bin/python

        echo "[backend]" >> inventory.ini
        echo "u21.local ansible_host=$BACKEND_IP ansible_user=ubuntu ansible_ssh_private_key_file=/var/jenkins_home/Linux-key.pem ansible_python_interpreter=/usr/bin/python3 
        """
      }
    }

    stage('Run Ansible') {
      steps {
        sh 'ansible-playbook -i inventory.ini playbook.yml'
      }
    }

```

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```
TASK [Start Netdata] ***********************************************************
ok: [u21.local]

PLAY RECAP *********************************************************************
c8.local                   : ok=7    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
u21.local                  : ok=5    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

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